

## Dataset (§4.2-4.3)

### Native GAIA tasks

*Text*

*Document*

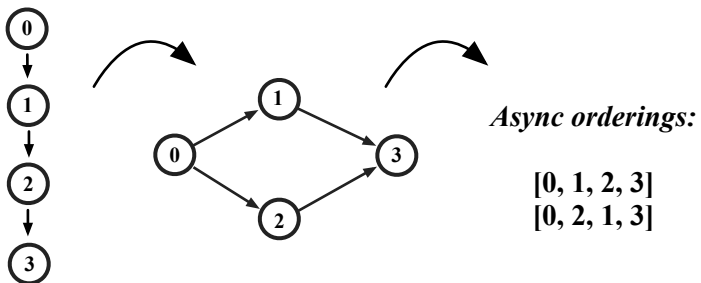
*Vision*

*Audio*

query · metadata · attachments · chain-style substeps.

### Added annotation layers [ours]

- ▶ **Tool layer.** 16-tool universe  $\tau$ ; per-domain visible subset  $\mathcal{T}_Q \subseteq \tau$ .
- ▶ **Gold dependency DAG.**  $G_{\text{plan}}^*(Q) = (\mathcal{P}_Q^*, \mathcal{D}_Q^*)$
- ▶ **Async ordering augmentation.** replay-validated linear orderings consistent with  $\mathcal{D}_Q^*$ .



## Inference pipeline (§4.1)

### STAGE 1 Planning

Planner reads  $Q$  and produces a planning graph over subgoals and dependencies.

**OUTPUT**  $\hat{G}_{\text{plan}} = (\hat{\mathcal{P}}_Q, \hat{\mathcal{D}}_Q)$



### STAGE 2 Execution

Iteratively call tools from  $\mathcal{T}_Q$  and collect observations.

**OUTPUT**  $\hat{\tau} = ((\hat{c}_1, o_1), \dots, (\hat{c}_T, o_T))$



### STAGE 3 Answer generation

Synthesize the final response from  $Q$  and  $\hat{\tau}$

**OUTPUT**  $y = f_{\text{answer}}(Q, \hat{\tau})$

## Evaluation (§4.5)

### (A) Plan structural fidelity

scored on  $\hat{G}_{\text{plan}}$  against augmented reference set

NodeF1 · EdgeF1 · NodeLabelSim

### (B) Tool invocation accuracy

scored on realized calls  $\hat{c}_t = (\text{name}_t, \text{args}_t)$ ,

ToolF1 · ParamF1 · ParamValueF1

### (C) End-task answer correctness

scored on  $y$  against gold answer

EM · TokenF1 · Accepted Rate